


```
data.isna().sum()
```

index	...	↑↓	0	...	↑↓
#				0	
claim_status				298	
video_id				0	
video_duration_sec				0	
video_transcription_text				298	
verified_status				0	
author_ban_status				0	
video_view_count				298	
video_like_count				298	
video_share_count				298	
video_download_count				298	
video_comment_count				298	
Rows: 12 ↗ Expand					

```
# Drop rows with missing values
data = data.dropna(axis=0)
```

```
data.head()
```

...	↑↓	...	↑↓	clai...	...	↑↓	vi...	...	↑↓	video_duration...	...	↑↓	video_transcription_text	...	↑↓	verified_st...	...	↑↓	author_ban_s...	...	↑↓	video_view...	...	↑↓	video_like_...	...	↑↓	video_share_...	...
	0		1	claim			7017666017					59	someone shared with me that drone deliveri...			not verified			under review				343296			19425			2
	1		2	claim			4014381136					32	someone shared with me that there are mor...			not verified			active				140877			77355			190
	2		3	claim			9859838091					31	someone shared with me that american indu...			not verified			active				902185			97690			28
	3		4	claim			1866847991					25	someone shared with me that the metro of s...			not verified			active				437506			239954			348
	4		5	claim			7105231098					19	someone shared with me that the number of...			not verified			active				56167			34987			41
Rows: 5																													

```
data.groupby("verified_status")["video_view_count"].mean()
```

verified_status	video_view_count
not verified	265663.7853388503
verified	91439.1641666667
Rows: 2	

Null hypothesis: There is no difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to chance or sampling variability).
Alternative hypothesis: There is a difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to an actual difference in the corresponding population means).

State the null hypothesis and the alternative hypothesis Choose a significance level Find the p-value Reject or fail to reject the null hypothesis

H_0 : There is no difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to chance or sampling variability).

H_A : There is a difference in number of views between TikTok videos posted by verified accounts and TikTok videos posted by unverified accounts (any observed difference in the sample data is due to an actual difference in the corresponding population means).

I choose 5% as the significance level

```
# Conduct a two-sample t-test to compare means
not_verified = data[data['verified_status'] == 'not verified']['video_view_count']
verified = data[data['verified_status'] == 'verified']['video_view_count']
stats.ttest_ind(a=not_verified, b=verified, equal_var=False)
```

```
TtestResult(statistic=np.float64(25.499441780633777), pvalue=np.float64(2.6088823687177823e-120), df=np.float64(1571.163074387424))
```

The analysis shows that there is a statistically significant difference in the average view counts between videos from verified accounts and videos from unverified accounts. This suggests there might be fundamental behavioral differences between these two groups of accounts.